

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Differentiation

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

DIFFERENTIATION · CH₃ · L10

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The Map

This lesson collapses Differentiation into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Differentiate powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find gradient at a point

TRIGGER *"work out", "find", a missing value*

03 Find stationary points

TRIGGER *"work out", "find", a missing value*

04 Use derivative in a context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Differentiate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 40 website questions, 40 checked solutions, 3 local PDFs read.

01 Differentiate powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$V = (2x + 5)(x + 1)(3 - x)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q15

The volume is

$$V = (2x + 5)(x + 1)(3 - x)$$

Expand brackets

$$(2x + 5)(x + 1) = 2x^2 + 7x + 5$$

Finish: $x = 1.47$ to 3 significant figures.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

15 (2x + 5) cm (x + 1) cm (3 - x) cm Diagram NOT accurately drawn The diagram shows a cuboid of volume $V \text{ cm}^3$ (a) Show that $V = 15 + 16x - x^2 - 2x^3$ (3) There is a value of x for which the volume of the cuboid is a maximum. (b) Find this value of x . Show your working clearly. Give your answer correct to 3 significant figures. $x = (5)$

YOUR SOLUTION

02 Find gradient at a point

TRIGGER *"work out", "find", a missing value*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = ax^3 + bx^2 - 12x + 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q24

Calculate value

$$y = ax^3 + bx^2 - 12x + 6$$

Point lies on the curve

$$-6 = 8a + 4b - 24 + 6$$

Finish: $y = 33$

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

24 The curve C has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants. The point A with coordinates (2, -6) lies on C. The gradient of the curve at A is 16. Find the y coordinate of the point on the curve whose x coordinate is 3. Show clear algebraic working.

y = O A S 100 A S

YOUR SOLUTION

03 Find stationary points

TRIGGER *"work out", "find", a missing value*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = x^3 - 8x^2 - 12x + 5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2024 P1HR Q19

Differentiate

$$y = x^3 - 8x^2 - 12x + 5$$

Stationary points have

$$3x^2 - 16x - 12 = 0$$

Finish: $A = (6, -139)$.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

19 A curve C has equation $y = x^3 - 2x^2 + 5x - 12$. Curve C has exactly two stationary points, one at point A and one at point B such that x coordinate of point A > x coordinate of point B. Find the coordinates of point A. Show clear algebraic working. (,)

YOUR SOLUTION

04 Use derivative in a context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q17

Volume of the solid is

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

The question gives

$$V = \frac{1}{3}\pi y$$

Finish: The maximum value of y is 320.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

17 A solid, S, is made from a hemisphere and a cylinder. The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point. x cm $(20 - 4x)$ cm Diagram NOT accurately drawn The radius of the cylinder and the radius of the hemisphere are both x cm. The height of the cylinder is $(20 - 4x)$ cm. The volume of S is V cm³ where $V = \frac{1}{3}\pi y$ Find the maximum value of y . Show clear algebraic working.

YOUR SOLUTION

05 Differentiate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q17

Volume of the solid is

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

The question gives

$$V = \frac{1}{3}\pi y$$

Finish: The maximum value of y is 320.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = x^2 - \frac{p}{x}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q22

Find the gradient

$$y = x^2 - \frac{p}{x}$$

Find the gradient

$$2(-3) + \frac{p}{(-3)^2} = 0$$

Finish: $p = 54$, $k = 12$

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative = 0.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

22 The diagram shows a sketch of part of the curve with equation $y = x^2 - px$ where p is a positive constant. Diagram NOT accurately drawn

y x T O

For all values of p , the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch. For the curve where the x coordinate of T is -3 (a) find the value of p $p = (4)$ The line with equation $y = k$ is a tangent to the curve with equation $y = x^2 - 16x$ (b) Find the value of k $k = (3)$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Differentiate powers
"work out", "find", a missing value	Find gradient at a point
"work out", "find", a missing value	Find stationary points
"work out", "show", "hence", a familiar exam phrase	Use derivative in a context
"work out", "show", "hence", a familiar exam phrase	Differentiate
"work out", "find", a missing value	Find the gradient

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.